

CSRMS

NADV 744 · Advanced Development Systems · Group Assignment 2026

Sol Plaatje University

(team names + student numbers)

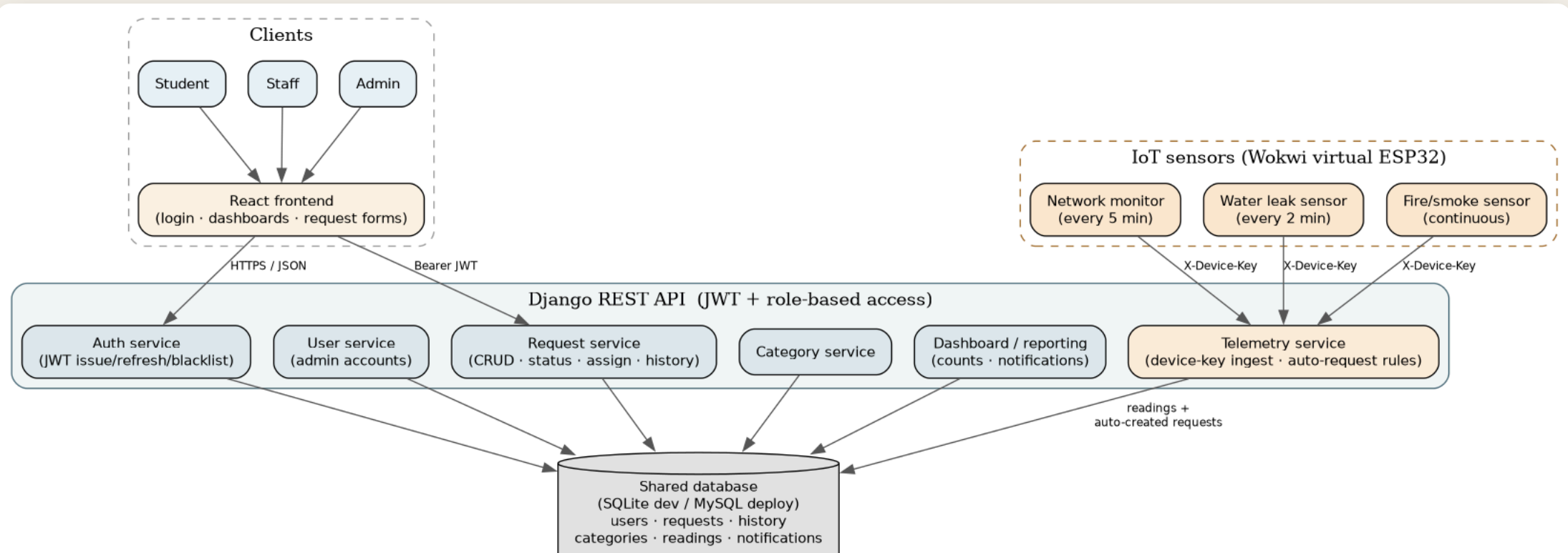
The problem

- Campus maintenance runs on WhatsApp and hallway conversations
- Reports arrive without structure: no category, no location, no owner
- Nobody can answer: *how many problems, how long to fix, which buildings?*
- Some failures **should not depend on a human noticing** — leaks and overheating happen at 2am

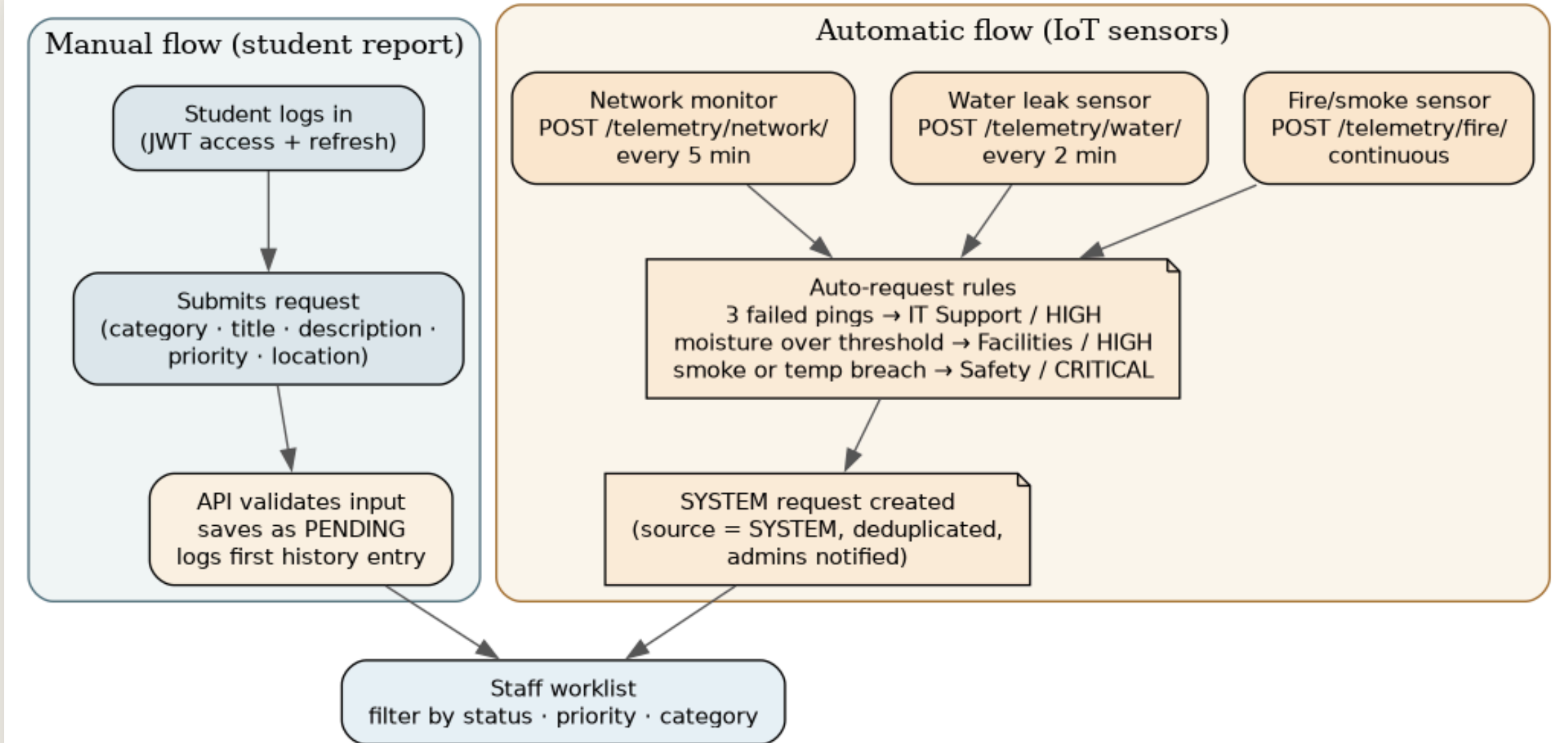
What we built

One system, two ways in:

- Students log problems → staff resolve them through a managed workflow
- Simulated IoT sensors raise requests **on their own** when conditions demand it



How a request travels



Security model

Layer	Decision
Auth	JWT on every endpoint (register/login/refresh excepted)
Roles	STUDENT / STAFF / ADMIN — registration always creates STUDENT, server-side
Objects	A student who guesses another's request ID gets a 404 , proven by tests
Devices	Per-sensor keys, SHA-256 hashed, type-bound, individually revocable
Passwords	Django validators + PBKDF2

IoT auto-request rules

Sensor	Cadence	Trigger	Ticket raised
Network monitor	5 min	3 consecutive failed pings	IT Support · HIGH
Water leak	2 min	moisture > 60%	Facilities · HIGH
Fire/smoke	10 s	smoke $\geq$ 40 or temp $\geq$ 50 °C	Safety · CRITICAL

Open SYSTEM tickets are **deduplicated** per sensor+location — a stuck sensor raises one ticket, not a hundred.

Live demo

Student report → staff workflow → sensor-triggered ticket

Student view

CSRMS
CAMPUS SERVICE DESK

Overview

Requests

SOL PLAATJE UNIVERSITY · STUDENT WORKSPACE

Hello, Naledi.

+ Log a request

YOUR SERVICE BRIEF

Track your campus requests.

Log a problem once and follow every step the service desk takes.

OPEN REQUESTS

1

IN PROGRESS

1

RESOLVED

1

MY TOTAL

3

Latest requests

CSR-1003

Low

Broken window pane in lecture hall B

Facilities · 12 min ago

Pending

CSR-1002

High

WiFi drops out every evening in Residence A

IT Support · 12 min ago

In progress

Staff workflow with full history

REQUEST DETAIL

CSR-1002

High

In progress

WiFi drops out every evening in Residence A

IT Support · Residence A common room · logged by Naledi Khumalo · 12 min ago

Reported via the campus portal. Please inspect Residence A common room.

SERVICE DESK ACTIONS

Mark resolved

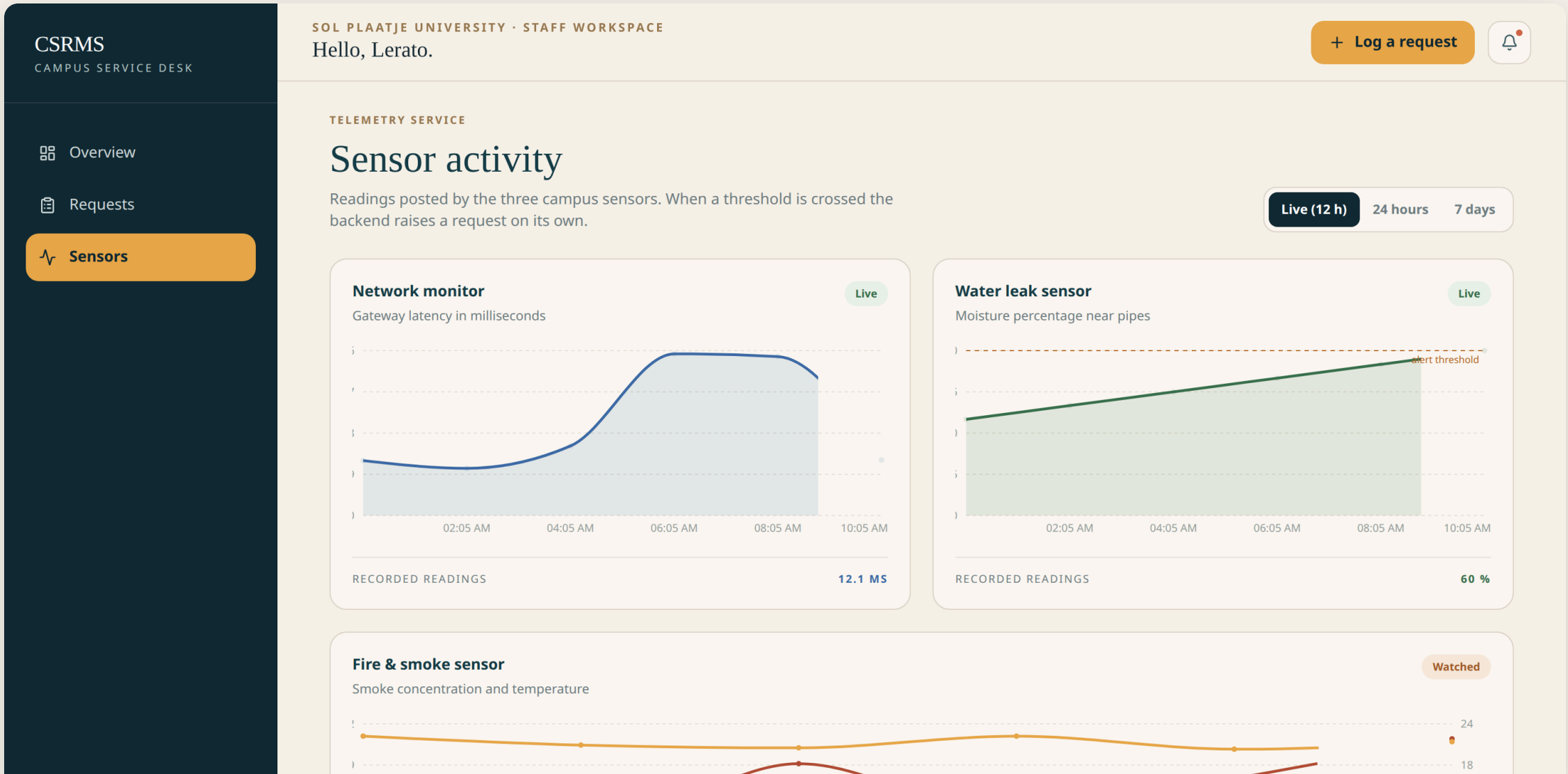
Reassign (now Lerato Mokoena)

Add a note to the timeline...

Post note

WORKFLOW HISTORY

Live sensor telemetry



Testing & evaluation

- **59 automated API/service tests** — every security claim has a negative test
- Firmware: structural validation + Wokwi scenarios driving sensors to their limits
- Playwright walkthrough of all three roles doubles as the screenshot generator
- Performance (50 rounds/endpoint): **p95 under 8 ms** on all reads;
JWT checks are DB-free, lists are paginated

Limitations & future work

Honest limits: simulated hardware · pull-only notifications · laptop-scale perf data

Next, in order:

1. Email/push notifications with staff digests
2. SLA tracking and automatic escalation
3. Real ESP32 hardware (same sketches, new WiFi credentials)
4. MySQL deployment behind HTTPS

Thank you

github.com/your-org/NADV74

`README.md` → running everything in three commands